

PRAKHAR BHANDARI

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[GitHub: Darelife](#) | [Codeforces: Darelife](#) | [Personal Website](#) | CGPA: 8.42

EDUCATION

BITS Pilani, Goa Campus

2023 – 2027

B.E. (Hons.) Computer Science · CGPA: 8.42

Relevant Coursework: Operating Systems, Computer Networks, Data Storage Technologies and Networks, Data Compression, Database Management Systems, Data Structures & Algorithms, Object Oriented Programming, Computer Programming

Languages: C++, Python, Java, Go, C, Bash

Systems: Linux, Docker, Kubernetes, GCP, CI/CD

Tools: Git, JJ, Postman

Backend & Data: PostgreSQL, MySQL, MongoDB, Redis

Frameworks: Flask, Express.js, Next.js, Firebase

Concepts: Data Compression, Distributed Systems, DBMS

WORK EXPERIENCE

Google | Software Engineering Intern | [google.com](#)

May 2026 – Aug 2026

- Worked on **Google Cloud Datastream**, a distributed serverless CDC service for real-time, heterogeneous-database replication.
- Built and automated a high-accuracy error-classification pipeline for **MySQL, PostgreSQL, MongoDB, SQL Server, and Oracle**, mapping **~250 errors** to originating services, eliminating false positives, and automating week-long rollouts.
- Protected a **7-hr/month error budget** by reducing unmapped errors and suppressing the team's primary paging source.
- Built an **on-call observability dashboard** correlating SLO fast burns with pod rollouts to diagnose multiple production outages.

AI Gurukul | Software Engineering Intern | [aigurukul.dev](#)

May 2025 – Jul 2025

- Developed AI agents using **CrewAI** and **Gemini API** with **PostgreSQL** for **XPIA** and **E-Library**.
- Implemented a PDF RAG pipeline combining **BM25 keyword matching** and **vector search** to reduce LLM hallucinations.
- Simulated an automated Cross Platform Injection Attack (**XPIA**) to manipulate a **Supabase Postgres DB**.

PROJECTS

PacMan: Network Packet Analyzer

Mar 2025

Full-Stack Development | [github.com/Darelife/PacMan-hackenza2.0](#)

- Won **2nd place out of 200+ teams** in a 24-hr hackathon building a web PCAP/PCAPNG network packet analyzer.
- Engineered a full-stack deep packet inspection platform using **Next.js, Flask, Scapy**, and **Pyshark** to capture live network packets, parse multi-layer protocol headers and payloads, and stream real-time telemetry metrics via WebSockets.

Codeforces Helper – Browser Extension

Dec 2024 – Apr 2025

Browser Extension Development

- Published browser extension on [Chrome Web Store](#) & [Firefox Add-ons](#) serving **2,000+ active users**.
- Engineered contest tracking, submission rating overlays, tag hiding filters, and custom problem list JSON backups, integrated with asynchronous **Codeforces REST API** sync and dynamic rating performance graphs.

Sing Assist – Real-Time Audio Analysis

Feb 2026 – Apr 2026

Systems Programming | [github.com/Darelife/singAssist](#)

- Developed a low-latency vocal pitch tracking engine in **Go** utilizing **FFT** and **pYIN** algorithms to extract audio frequencies.
- Designed real-time audio signal processing with **moving-average pitch smoothing**, octave-jump suppression, and frequency tolerance matching algorithms to compare user vocal pitch against reference audio tracks and render live precision scores.

Predictive Prefetching for LLMs (Marvell Technologies)

Jan 2026 – Apr 2026

Machine Learning | [github.com/Darelife/Adaptive-Predictive-Caching-for-RL-Workloads](#)

- Designed a 3-tier predictive caching system (HBM → DRAM → NVMe) for AI/ML training, combining a **SHADE-inspired heuristic**, an **XGBoost predictor**, and a **contextual bandit** to prefetch “hot” data before the GPU needs it.
- Validated on real access traces from live **PyTorch ResNet/CIFAR-10** training runs using a custom simulator, achieving a **29.98% cache hit rate** and cutting I/O latency by **~28% (300ms → 216ms)** vs. an LRU baseline.

AWARDS & RECOGNITIONS

Expert on Codeforces (Peak Rating: **1643**) | Handle: [darelife](#)

Nov 2023 – Present

- Solved **750+ competitive programming problems** and competed in **65+ contests**.

LEADERSHIP & EXPERIENCE

AlgoManiaX (Competitive Programming Club, BITS Goa) | Coordinator

Mar 2024 – Present

- Led a **70+ member** club; built the [AlgoManiaX website](#) and hosted its [Discord bot](#) on a **Google Cloud VM**.
- Organized **40+ events** for **1,000+ unique participants** and created **Manim** animations for problem explanations.